Invoice Extraction AI

Objective

Build an AI-powered application that extracts structured data from invoice documents (images or PDFs), stores results in a database, and enables analytics on extracted data.

Input

- Upload JPG / PNG / PDF invoice files

Core Requirements

1. File Upload

Simple UI & API endpoint. Accept multiple file formats.

2. OCR Processing

Extract raw text from document using Tesseract or external APIs.

3. LLM Parsing Layer

Convert raw OCR text into structured JSON. Handle different invoice formats, missing fields, and noisy OCR output.

4. Output Validation

Ensure valid JSON and correct data types. Handle errors gracefully.

5. Deployment

Deploy the app and provide a live URL or API endpoint.

Requirements

1. Supabase Integration

Use Supabase as the backend database.
- Store users
- Store uploaded files metadata
- Store extracted invoice data

2. File Storage

Store uploaded invoice files in Supabase Storage.
Save file URL/path in database.

3. Format Detection & Reuse

If a similar invoice format is uploaded again:
- Detect format similarity
- Reuse parsing logic/template
- Improve extraction accuracy and speed

5. Analytics Layer

Build basic analytics on extracted data:
- Total spend by vendor
- Monthly spend trends
- Number of invoices processed
- Currency-wise totals

Tech Stack

Backend: FastAPI / Flask
Frontend: React
OCR
LLM: OpenAI / Gemini
Deployment: Vercel / Render

Bonus Features

- Multi-invoice batch processing
- Confidence score
- Field highlighting
- Retry logic
- Vendor normalization

Evaluation Criteria

AI/ML Thinking (30%)
Engineering Quality (25%)
Robustness (20%)
Product Thinking (15%)
Deployment (10%)

Evaluation Additions

Database Design (15%)
Scalability Thinking (10%)
Analytics & Insights (10%)

Feature

- Auto-group invoices by vendor
- Dashboard for analytics
- Detect duplicate invoices
- Learning system that improves over time

Submission Requirements

Please ensure your submission includes:

- **Codebase:** Public GitHub repository with clear structure
- **Live Demo:** Deployed application link (web app or API)
- **Test Data:** Sample invoices used for validation
- **Documentation (README):**
    - System architecture
    - Key design decisions
    - Assumptions & limitations
    - Potential improvements
- **Demo Video:** 2–5 minute walkthrough explaining your solution
- **Email : yashjeet.singh@avaipl.com**

Time Limit

48 hours

What Will Impress Us

Strong prompt engineering, fallback mechanisms, clean validation, cost optimization, and edge case handling.